Now, you might not think this is particularly surprising. Water is made of H_2O molecules, and a water wave arises when many molecules move in a coordinated pattern. One group of H_2O molecules goes up in one location, while another group goes down in a nearby location. Perhaps the data illustrated in Figure 4.3 show that electrons, like H_2O molecules, sometimes move in concert, creating a wavelike pattern in their overall, macroscopic motion. While at first blush this might seem to be a reasonable suggestion, the actual story is far more unexpected.

We initially imagined that a flood of electrons was fired continuously from the electron gun in Figure 4.3. But we can tune the gun so that it fires fewer and fewer electrons every second; in fact, we can tune it all the way down so that it fires, say, only one electron every ten seconds. With enough patience, we can run this experiment over a long period of time and record the impact position of each individual electron that passes through the slits. Figures 4.4a–4.4c show the resulting cumulative data after an hour, half a day, and a full day. In the 1920s, images like these rocked the foundations of physics. *We see that even individual, particulate electrons, moving to the screen independently, separately, one by one, build up the interference pattern characteristic of waves.*

This is as if an individual H_2O molecule could still embody something akin to a water wave. But how in the world could that be? Wave motion seems to be a collective property that has no meaning when applied to separate, particulate ingredients. If every few minutes individual spectators in the bleachers get up and sit down separately, independently, they are *not* doing the wave. More than that, wave interference seems to require a wave from *here* to cross a wave from *there*. So how can interference be at all relevant to single, individual, particulate ingredients? But somehow, as attested by the interference data in Figure 4.4, even though individual electrons are tiny particles of matter, each and every one also embodies a wavelike character.